

Parametric Equations

ANSWER KEY



Derivatives of parametric equations

- 1 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{dy}{dx}$ in terms of t .
$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{3t^2}{2t} = \frac{3t}{2}.$$
- 2 For $x(t) = \cos t$, $y(t) = \sin t$, find $\frac{dy}{dx}$ and evaluate it at $t = \frac{\pi}{4}$.
$$\frac{dy}{dx} = \frac{\cos t}{-\sin t} = -\cot t. \text{ At } t = \frac{\pi}{4}: -\cot\left(\frac{\pi}{4}\right) = -1.$$

Second derivatives of parametric equations

- 3 For $x(t) = t^2$, $y(t) = t^3$, find $\frac{d^2y}{dx^2}$ using $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$.
$$\frac{dy}{dx} = \frac{3t}{2}, \frac{d}{dt}\left(\frac{3t}{2}\right) = \frac{3}{2}, \frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t}.$$

Tangent lines to parametric curves

- 4 Find the equation of the tangent line to the curve $x(t) = t^2 - 1$, $y(t) = t^3 - t$ at $t = 2$.
At $t = 2$: $x = 3$, $y = 6$. $\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 1 = 11$. $\frac{dy}{dx} = \frac{11}{4}$. Tangent: $y - 6 = \frac{11}{4}(x - 3)$.

Arc length of parametric curves

- 5 State the arc length formula for a parametric curve $x(t)$, $y(t)$ on $[a, b]$.
$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$
- 6 Find the arc length of $x(t) = 3t$, $y(t) = 4t$ for $t \in [0, 2]$.
$$\frac{dx}{dt} = 3, \frac{dy}{dt} = 4. L = \int_0^2 \sqrt{9 + 16} dt = \int_0^2 5 dt = 10.$$
- 7 Set up (but do not evaluate) the arc length integral for $x(t) = t^2$, $y(t) = t^3$ on $[0, 1]$.
$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 3t^2. L = \int_0^1 \sqrt{(2t)^2 + (3t^2)^2} dt = \int_0^1 \sqrt{4t^2 + 9t^4} dt.$$

Speed as a vector quantity

- 8 A particle moves with position $x(t) = t^2$, $y(t) = 2t$. Find its speed at $t = 1$.
$$\frac{dx}{dt} = 2t, \frac{dy}{dt} = 2. \text{ At } t = 1: \frac{dx}{dt} = 2, \frac{dy}{dt} = 2. \text{ Speed} = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}.$$

- 9 A particle moves with position $x(t) = 3\cos t$, $y(t) = 3\sin t$. Find its speed at any time t , and interpret the result.
- $\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. Speed $= \sqrt{9\sin^2 t + 9\cos^2 t} = \sqrt{9} = 3$ (constant). The particle traces a circle of radius 3 at constant speed.
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Eliminating the parameter

- 10 Eliminate the parameter from $x(t) = t$, $y(t) = t^2 - 1$ to find the Cartesian equation, and sketch it.
Since $x = t$, substitute directly: $y = x^2 - 1$, an upward parabola shifted down 1 unit.
- 11 Eliminate the parameter from $x(t) = 2t + 1$, $y(t) = 4t - 3$ to find the Cartesian equation.
From $x = 2t + 1$: $t = \frac{x-1}{2}$. Substitute: $y = 4\left(\frac{x-1}{2}\right) - 3 = 2(x-1) - 3 = 2x - 5$. A line with slope 2.
- 12 Explain why eliminating the parameter from $x(t) = \cos t$, $y(t) = \sin t$ using the identity $\sin^2 t + \cos^2 t = 1$ gives a relation that is NOT a function $y = f(x)$.
Substituting gives $x^2 + y^2 = 1$, the equation of a unit circle. For most x -values in $(-1, 1)$, there are two corresponding y -values ($\pm\sqrt{1-x^2}$), so this relation fails the vertical line test and is not a function of x .
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Concavity of parametric curves

- 13 For $x(t) = t^2$, $y(t) = t^3$ (with $\frac{d^2y}{dx^2} = \frac{3}{4t}$ found earlier), determine the concavity of the curve for $t > 0$ and for $t < 0$.
For $t > 0$: $\frac{d^2y}{dx^2} = \frac{3}{4t} > 0$, so the curve is concave up. For $t < 0$: $\frac{3}{4t} < 0$, so the curve is concave down.
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Horizontal and vertical tangents

- 14 For $x(t) = t^2 - 4$, $y(t) = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal (where $\frac{dy}{dt} = 0$, provided $\frac{dx}{dt} \neq 0$).
 $\frac{dy}{dt} = 3t^2 - 3 = 0 \Rightarrow t = \pm 1$. Check $\frac{dx}{dt} = 2t$: at $t = \pm 1$, $\frac{dx}{dt} = \pm 2 \neq 0$, so both give valid horizontal tangents.
- 15 For the same curve, find the value(s) of t where the tangent line is vertical (where $\frac{dx}{dt} = 0$, provided $\frac{dy}{dt} \neq 0$).
 $\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. Check $\frac{dy}{dt} = 3(0)^2 - 3 = -3 \neq 0$, so $t = 0$ gives a valid vertical tangent, at the point $x(0) = -4$, $y(0) = 0$.
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Motion along a parametric path (particle interpretation)

- 16** A particle's position is $x(t) = t^2$, $y(t) = t^3 - 3t$ for $t \in [-2, 2]$. Find all times when the particle is moving straight up or down (vertical velocity component only, i.e. $\frac{dx}{dt} = 0$).
 $\frac{dx}{dt} = 2t = 0 \Rightarrow t = 0$. At $t = 0$: position is $(0, 0)$, and the particle is momentarily moving purely vertically (since horizontal velocity is 0 there).
- 17** For the particle above, find the speed at $t = 2$.
 $\frac{dx}{dt} = 2t = 4$; $\frac{dy}{dt} = 3t^2 - 3 = 9$. Speed $= \sqrt{4^2 + 9^2} = \sqrt{16 + 81} = \sqrt{97} \approx 9.85$.

Arc length of a full circle (verifying the circumference formula)

- 18** Use the parametric arc length formula to verify that the circumference of a circle $x(t) = r \cos t$, $y(t) = r \sin t$ for $t \in [0, 2\pi]$ is $2\pi r$.
 $\frac{dx}{dt} = -r \sin t$, $\frac{dy}{dt} = r \cos t$. $L = \int_0^{2\pi} \sqrt{r^2 \sin^2 t + r^2 \cos^2 t} dt = \int_0^{2\pi} r dt = r(2\pi) = 2\pi r$, matching the known circumference formula.
- 19** Find the arc length of the parametric curve $x(t) = e^t \cos t$, $y(t) = e^t \sin t$ is a spiral; set up (but do not evaluate) its arc length integral on $[0, \pi]$.
 $\frac{dx}{dt} = e^t \cos t - e^t \sin t$, $\frac{dy}{dt} = e^t \sin t + e^t \cos t$. After simplification, $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 2e^{2t}$. So
 $L = \int_0^\pi \sqrt{2e^{2t}} dt = \int_0^\pi e^t \sqrt{2} dt$.

Parametric equations from real motion

- 20** A Ferris wheel of radius 10 m completes one revolution every 40 s, centered 12 m above the ground. Write parametric equations for a rider's height above the ground as a function of time t (seconds), assuming the rider starts at the bottom.
Angular speed $= \frac{2\pi}{40} = \frac{\pi}{20}$ rad/s. Starting at the bottom means height $= 12 - 10 \cos\left(\frac{\pi}{20}t\right)$ (using $x(t) = t$ for time, $y(t) = 12 - 10 \cos\left(\frac{\pi}{20}t\right)$ for height).
- 21** For the Ferris wheel model above, find the rider's vertical velocity $\frac{dy}{dt}$ at $t = 10$ s (a quarter revolution).
 $\frac{dy}{dt} = 10 \cdot \frac{\pi}{20} \sin\left(\frac{\pi}{20}t\right) = \frac{\pi}{2} \sin\left(\frac{\pi}{20}t\right)$. At $t = 10$: $\frac{\pi}{2} \sin\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \approx 1.57$ m/s.